Ohm’s law

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Connect voltage, current and resistance for an ohmic component.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 3.*
Positive charge: left. Negative charge: right. Each force has magnitude 15 N.

02. Worked example: explain the model
Closed loop; 12 V across 4 ohms; 3 A conventional current from positive supply terminal through resistor.

03. Read and interpret the model
Straight line through origin; slope V/I is resistance.

04. Apply the physics
2 A.

05. Explain your reasoning
Current halves because I=V/R.

06. Review: connect an earlier idea *Review: Unit 6, Day 12.*
Same speed c; the higher-frequency wavelength is 1/4 as large.

07. Exit check
8 V.

08. STRETCH: test the model
E.g.(4 V, 2 ohms),(8 V, 4 ohms),(12 V, 6 ohms). To isolate resistance, hold voltage fixed and vary resistance.

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01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 3.*
Field becomes 1/9; assumes an isolated point charge or an exterior spherically symmetric source.

02. Worked example: explain the model
Closed loop; 12 V across 4 ohms; 3 A conventional current from positive supply terminal through resistor.

03. Read and interpret the model
Straight line through origin; slope V/I is resistance.

04. Apply the physics
30 Ω; 0.30 A.

05. Explain your reasoning
Current halves because I=V/R.

06. Review: connect an earlier idea *Review: Unit 6, Day 12.*
λ≈0.05 m; microwave.

07. Exit check
20 Ω; V/I is constant for these measurements.

08. STRETCH: test the model
R 8 ohms; I 1.25 A. Temperature/material changes can alter resistance.